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Course Name: SIGNALS AND SYSTEM

Course Code: ECA 0304

Experiment -18: Z-Transform, ROC and Inverse Z-Transform of Discrete-Time Signals

OBJECTIVES

• To compute the Z-transform of a selected discrete-time signal using MATLAB.

• To determine the inverse Z-transform and recover the corresponding time-domain sequence.

• To identify the ROC and visualize pole-zero locations.

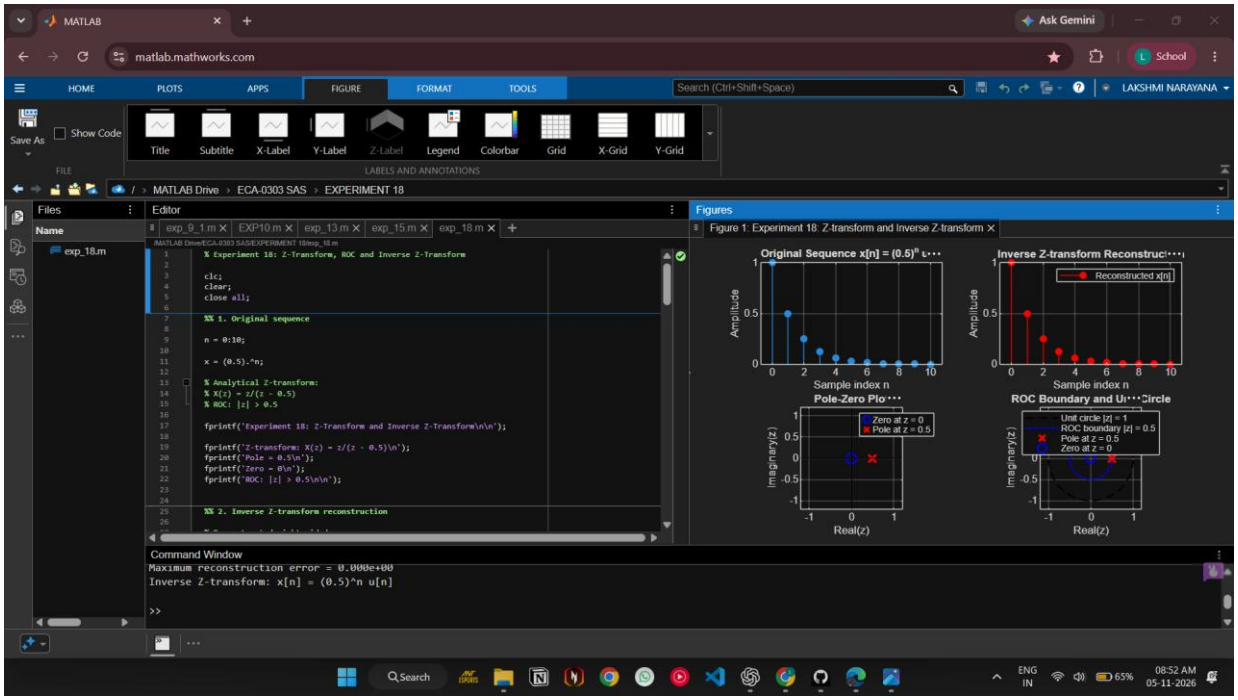
• To compare the original and reconstructed sequences and interpret the relationship between the ROC and stability.

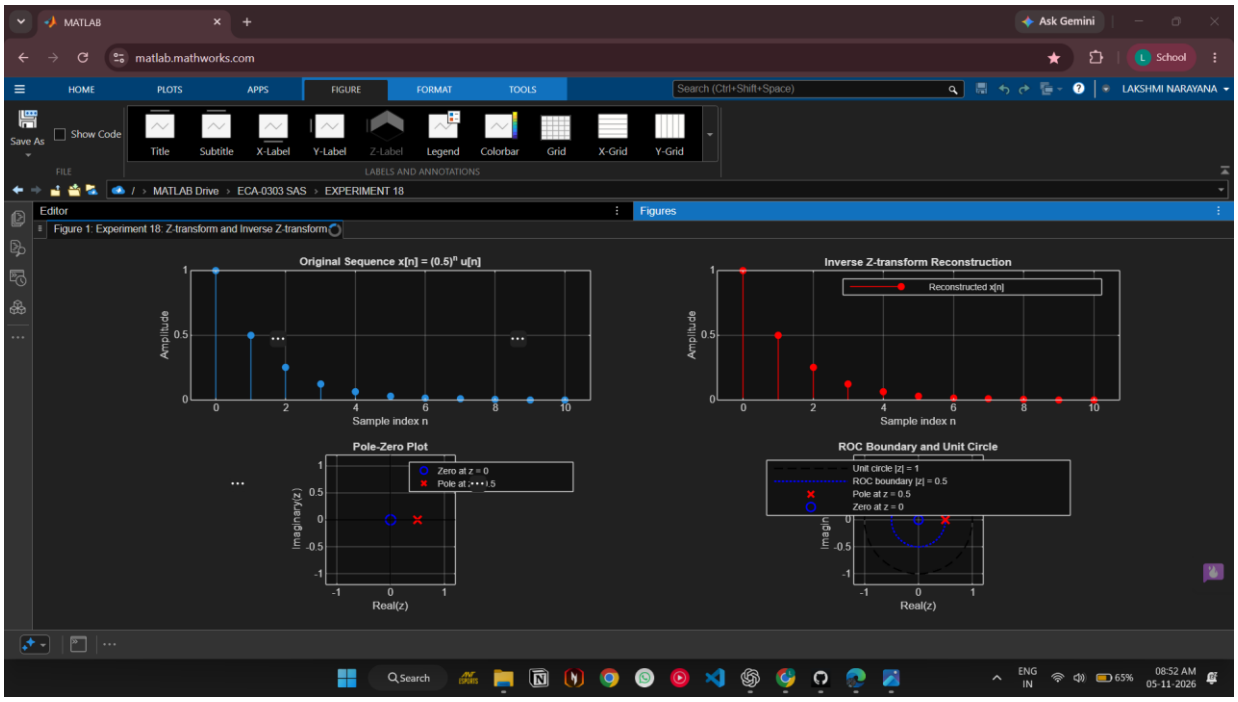
MATLAB Coding

```
% Experiment 18: Z-Transform, ROC and Inverse Z-Transform
clc; clear; close all;
% x[n] = (0.5)^n u[n], a right-sided sequence
n = 0:10;
x = (0.5).^n;
% Analytical Z-transform: X(z) = z/(z - 0.5)
% ROC for the right-sided infinite sequence: |z| > 0.5
syms z k
Xz = z/(z - 0.5);
xInverse = iztrans(Xz, z, k);
disp('Z-transform X(z) = ');
pretty(Xz);
disp('Inverse Z-transform (symbolic) = ');
pretty(xInverse);
% Numerical inverse sequence for comparison over n = 0:10
xRec = (0.5).^n;
fprintf('Maximum reconstruction error = %.3e\n', max(abs(x-xRec)));
fprintf('Pole = 0.5; zero = 0; ROC for right-sided sequence: |z| > 0.5\n');
```

```
figure('Name','Experiment 18: Z-transform and inverse');
subplot(2,2,1);
stem(n,x,'filled'); grid on;
xlabel('Sample index n'); ylabel('Amplitude');
title('Original sequence x[n] = (0.5)^n u[n]');
subplot(2,2,2);
stem(n,xRec,'r','filled'); grid on;
xlabel('Sample index n'); ylabel('Amplitude');
title('Inverse Z-transform reconstruction');
legend('Reconstructed x[n]');
subplot(2,2,3);
zplane([1 0],[1 -0.5]); grid on;
title('Pole-zero plot: zero at 0, pole at 0.5');
subplot(2,2,4);
theta = linspace(0,2*pi,500);
plot(cos(theta),sin(theta),'k--','LineWidth',1); hold on;
plot(0.5*cos(theta),0.5*sin(theta),'b:','LineWidth',1.2);
plot(0.5,0,'rx','MarkerSize',10,'LineWidth',2);
axis equal; grid on;
xlabel('Real(z)'); ylabel('Imag(z)');
title('ROC boundary |z| = 0.5 and unit circle');
legend('Unit circle |z|=1','ROC boundary','Pole at z=0.5','Location','best');
```

Input and Output Graphs


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Z-Label
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X-Grid
Y-Grid
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MATLAB Drive › ECA-0303 SAS › EXPERIMENT 18
Files
Name
exp_18.m
Editor
exp_9_1.m
EXP10.m
exp_13.m
exp_15.m
exp_18.m
% Experiment 18: Z-Transform, ROC and Inverse Z-Transform
clc;
clear;
close all;
%% 1. Original sequence
n = 0:10;
x = (0.5).^n;
% Analytical Z-transform:
% X(z) = z/(z - 0.5)
% ROC: |z| > 0.5
fprintf('Experiment 18: Z-Transform and Inverse Z-Transform\n\n');
fprintf('Z-transform: X(z) = z/(z - 0.5)\n');
fprintf('Pole = 0.5\n');
fprintf('Zero = 0\n');
fprintf('ROC: |z| > 0.5\n\n');
%% 2. Inverse Z-transform reconstruction
Figures
Figure 1: Experiment 18: Z-transform and Inverse Z-transform
Original Sequence x[n] = (0.5)^n u[n]
Amplitude
Sample index n
Inverse Z-transform Reconstruction
Reconstructed x[n]
Pole-Zero Plot
Zero at z = 0
Pole at z = 0.5
Imaginary(z)
Real(z)
ROC Boundary and Unit Circle
Unit circle |z| = 1
ROC boundary |z| = 0.5
Pole at z = 0.5
Zero at z = 0
Command Window
Maximum reconstruction error = 0.000e+00
Inverse Z-transform: x[n] = (0.5)^n u[n]
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Reconstructed x[n]
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Zero at z = 0
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Imaginary(z)
Real(z)
ROC Boundary and Unit Circle
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ROC boundary |z| = 0.5
Pole at z = 0.5
Zero at z = 0
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